

EchoMotion: Enhancing Exercise Analysis with Acoustic Sensing



Mohammed Mosuily, Jagmohan Chauhan
University of Southampton, United Kingdom
mtm1g19@soton.ac.uk, j.chauhan@soton.ac.uk

IntellEcT Systems
Lab

1 Introduction

Cardiovascular diseases remain a leading cause of death worldwide, and cardio exercise is essential for prevention. Remote exercise monitoring can provide valuable feedback for individuals, trainers, and clinicians, but existing solutions have notable limitations. Vision-based systems compromise privacy and depend heavily on lighting, wearable devices can be intrusive and inconvenient, and RF/WiFi approaches require complex, expensive hardware. EchoMotion addresses these challenges by introducing the first inaudible acoustic-based smart speaker system for detailed form analysis and 3D pose estimation during cardio exercises. The system emits ultrasonic tones between 17–21 kHz, captures their reflections using a microphone array, and processes these signals through a deep learning model with a custom loss function. This enables sub-millimetre accuracy even for fast movements, without sacrificing privacy or requiring costly specialised equipment.

5 EchoMotion System

The EchoMotion pipeline begins with ultrasonic tone emission, followed by capturing reflections via the microphone array. Two key features are extracted from the recorded signals: the Mel-spectrogram, which provides a perceptually motivated frequency representation, and spectral contrast, which captures differences between spectral peaks and valleys. These features are fed into a deep learning model consisting of five dense layers with batch normalization and dropout, optimised with a custom weighted joint loss that prioritises accuracy for critical joints such as elbows and wrists. The system outputs the 3D coordinates of 13 key joints, including the pelvis, shoulders, elbows, wrists, hips, knees, and ankles. The architecture is designed to be efficient, enabling high accuracy without requiring high-performance computing hardware.

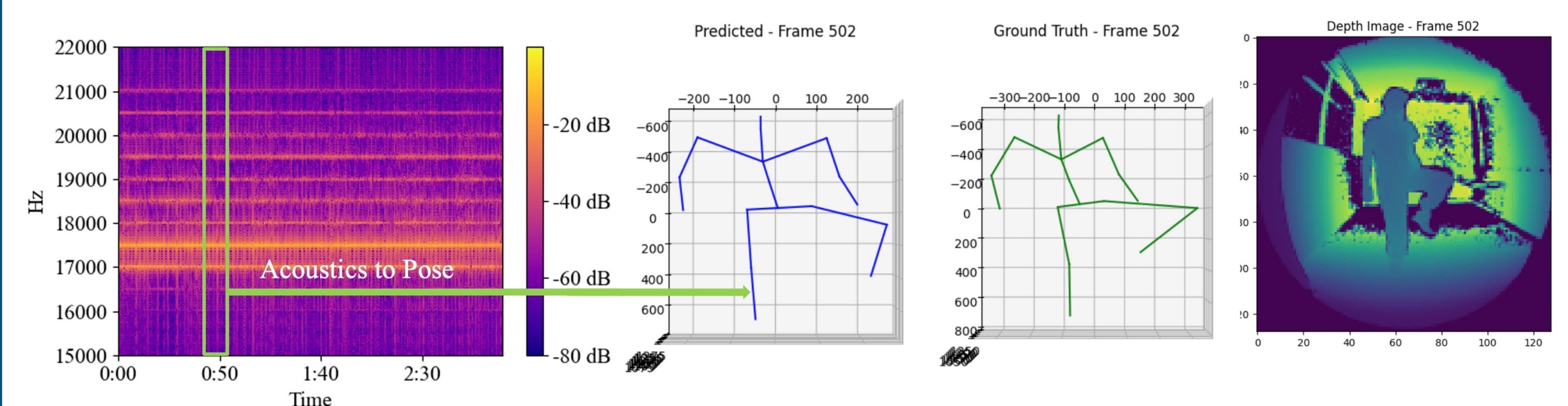


Figure: EchoMotion: One Frame of High Knee Exercise

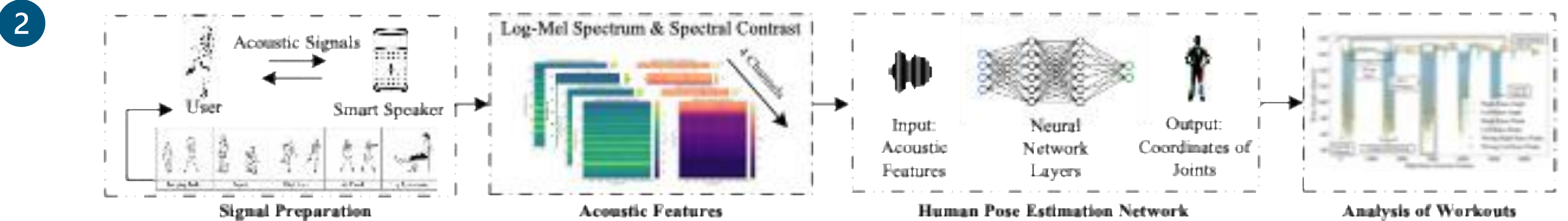


Figure: Flowchart representing acoustic signal analysis and processing pipeline for EchoMotion

3 Research Question

This study investigates whether inaudible acoustic signals can be leveraged to achieve high-accuracy 3D human pose estimation for fast cardio exercises in home-like environments, while preserving privacy and keeping hardware costs low.

4 Dataset

The dataset was collected from 22 participants aged 18–44, representing seven ethnic backgrounds, with heights ranging from 164 to 187 cm, weights from 50 to 95 kg, and BMIs from 18.3 to 32.4. Exercises were performed in two settings: a living room (6.62×3.6 m) for 13 participants, and a laboratory (6×2.7 m) for 9 participants.

The recording setup included a MiniDSP UMA-8 USB microphone array with 7 microphones, a Bose Companion 2 speaker for tone generation, and an Azure Kinect DK for collecting ground truth 3D pose data. The dataset totals approximately 11 hours of recordings, producing around 880,000 frames. Ultrasonic tones at 17–21 kHz were recorded at a 44.1 kHz sample rate. Controlled background noise conditions of 40 dB, 60 dB, and 80 dB were introduced to test robustness.

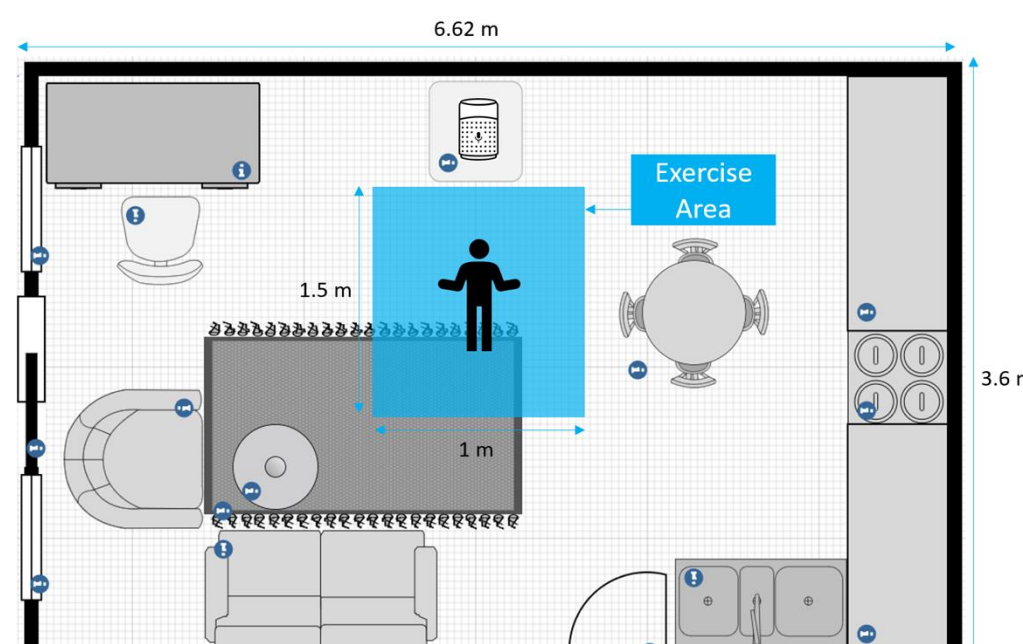


Figure : Data Collection in the living room

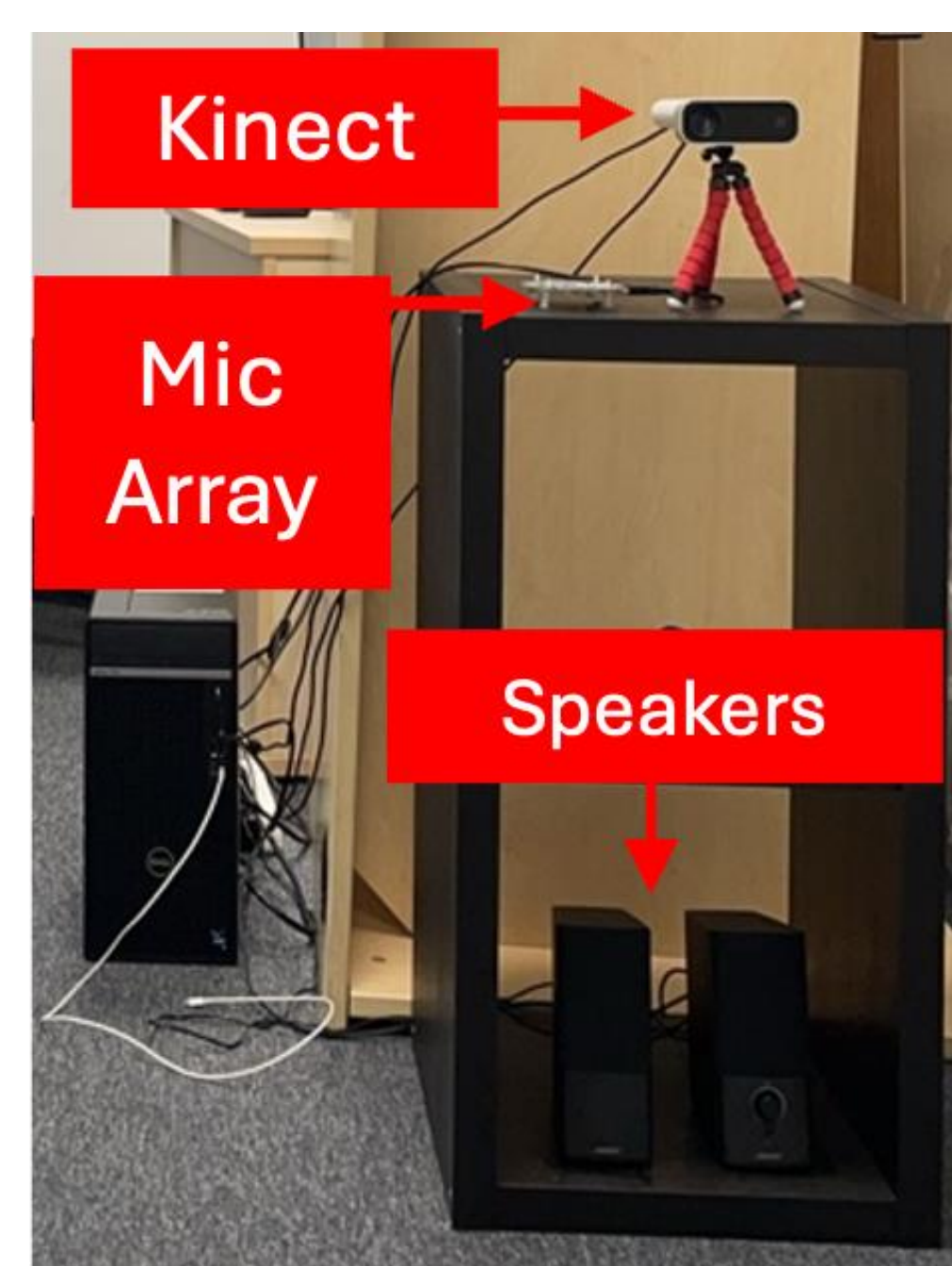


Figure: Hardware setup: Kinect, mic array, and speaker positions.

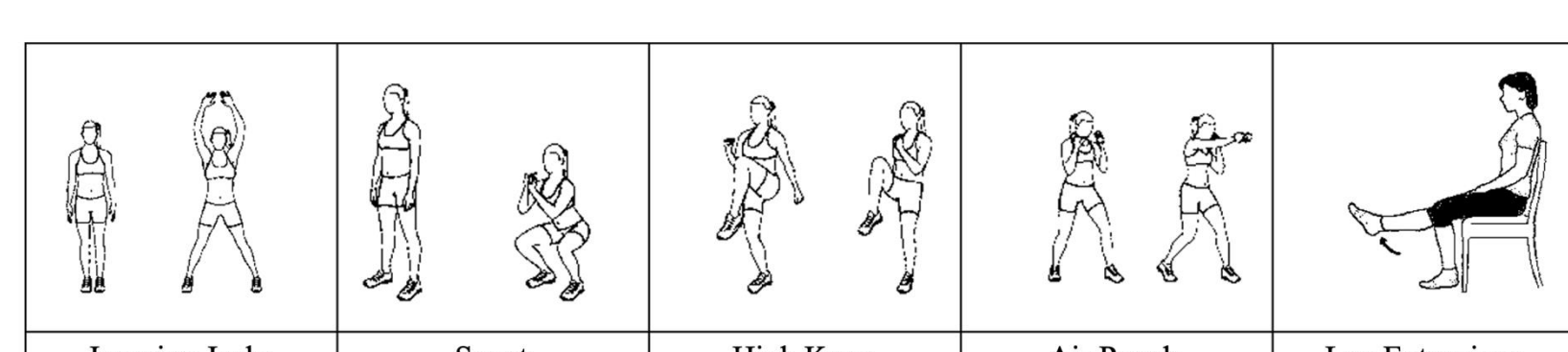


Figure: List of Exercises

6 Results

In single-subject experiments, EchoMotion achieved an average Mean Absolute Error (MAE) of 0.59 mm across all exercises, significantly outperforming reported values of 2.8–96 mm from other state-of-the-art systems. Performance was consistent across environments: for example, Leg Extensions achieved MAEs of 0.55 mm in the living room and 0.50 mm in the lab, while Jumping Jacks achieved 0.58 mm and 0.59 mm respectively. In cross-subject experiments using a leave-one-out approach, MAEs ranged from 0.61 mm to 1.28 mm in the living room, and 0.91 mm to 2.71 mm in the lab, with the highest error occurring in Air Punch for the lab setting. EchoMotion also proved robust to noise and distance, maintaining an MAE of approximately 0.6 mm at 40 dB, 60 dB, and 80 dB, and at both 1 m and 2 m distances.

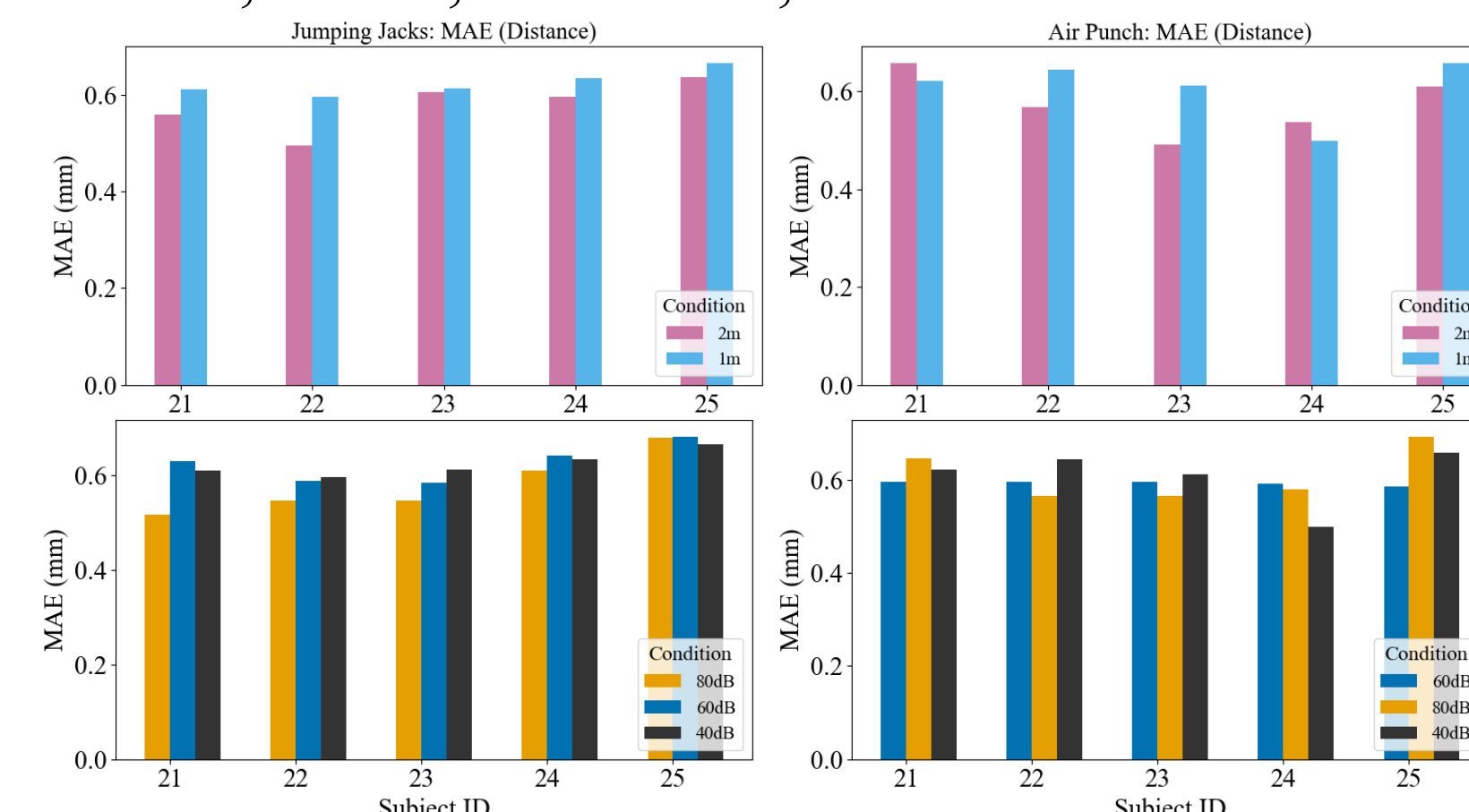


Figure: Impact of distance (above) and noise (below)

Table : comparison of EchoMotion with reported performance and Sensing techniques for pose estimation by movement speed

Method	Modality	MAE (mm)
GoPose	RF/WiFi	47
BodyTrak	RGB	69
PoseSonic	Audio	61
Shibata et al.	Audio	2.8
EchoMotion	Acoustic	0.59

7 Conclusion

EchoMotion demonstrates that inaudible acoustic sensing can deliver sub-millimetre accuracy in human pose estimation for fast cardio exercises, while preserving privacy, using low-cost hardware, and operating robustly across environments, noise levels, and distances. By making the dataset and code publicly available, we aim to encourage further research in acoustic-based exercise monitoring. Future work will focus on improving cross-subject generalisation and enabling real-time exercise feedback.